

1 Visual diagrams for Problems 9–10

This section collects diagrams and charts illustrating the two key mechanisms used in Problems 9–10:

$$\langle a, \partial x \rangle = \langle \delta a, x \rangle$$

and

$$f_* \beta_X = \beta_Y f_*.$$

The figures are intended to be inserted after the theory section or after the examples section.

1.1 Pairing and boundary for (D^{k+1}, S^k)

The pair

$$(D^{k+1}, S^k) = (D^{k+1}, \partial D^{k+1})$$

is the cleanest geometric model for the compatibility identity

$$\langle a, \partial x \rangle = \langle \delta a, x \rangle.$$

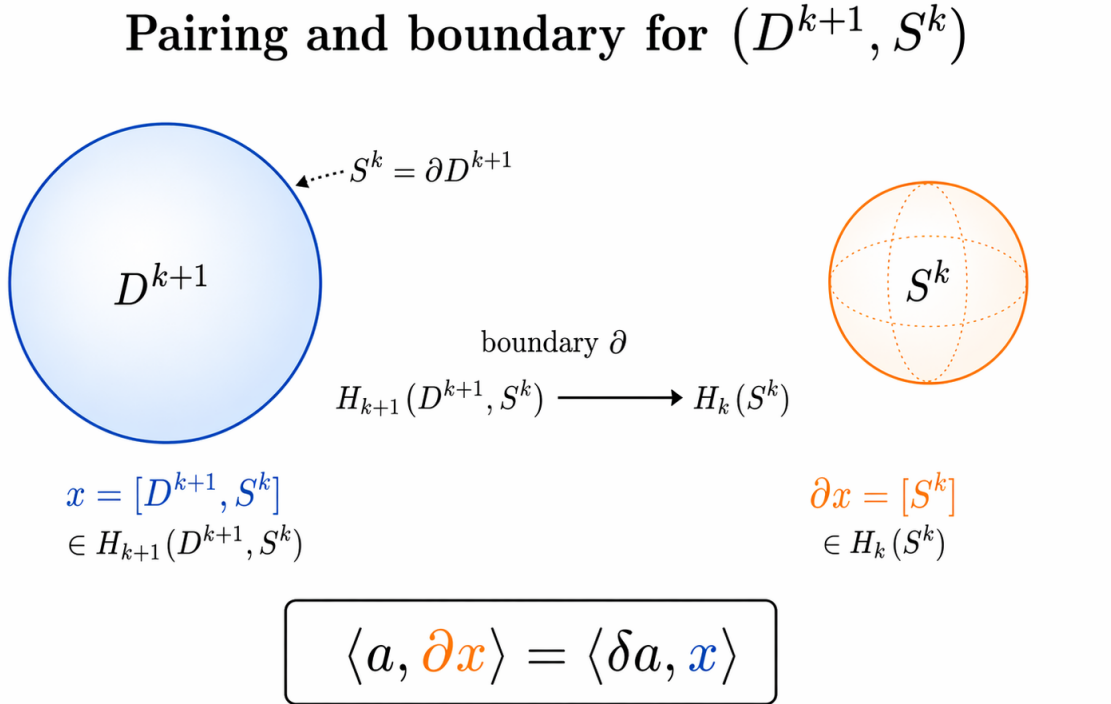
The relative class

$$x = [D^{k+1}, S^k] \in H_{k+1}(D^{k+1}, S^k)$$

has boundary

$$\partial x = [S^k] \in H_k(S^k).$$

Therefore evaluating a class on the boundary sphere is the same as evaluating the connecting cohomology class on the relative disk class.



Relative Stokes principle: evaluate on the boundary = evaluate the coboundary on the interior.

Figure 1: The relative Stokes principle for the pair (D^{k+1}, S^k) . The identity $\langle a, \partial x \rangle = \langle \delta a, x \rangle$ says that the homology boundary and the cohomology connecting map are adjoint under the Kronecker pairing.

1.2 Annulus example and the obstruction map $\delta(m, n) = m - n$

Let

$$X = S^1 \times [0, 1], \quad A = \partial X = S^1 \times \{0\} \sqcup S^1 \times \{1\}.$$

The boundary of the relative fundamental class is

$$\partial[X, A] = [S^1 \times \{1\}] - [S^1 \times \{0\}].$$

Since

$$H^1(A) \cong H^1(S^1 \sqcup S^1) \cong \mathbb{Z} \oplus \mathbb{Z},$$

a class on the boundary may be written as a pair

$$(m, n),$$

where the two entries record the values on the two boundary components. The connecting homomorphism measures the failure of these values to agree:

$$\delta(m, n) = m - n.$$

Thus the annulus gives a very concrete obstruction interpretation of the connecting homomorphism.

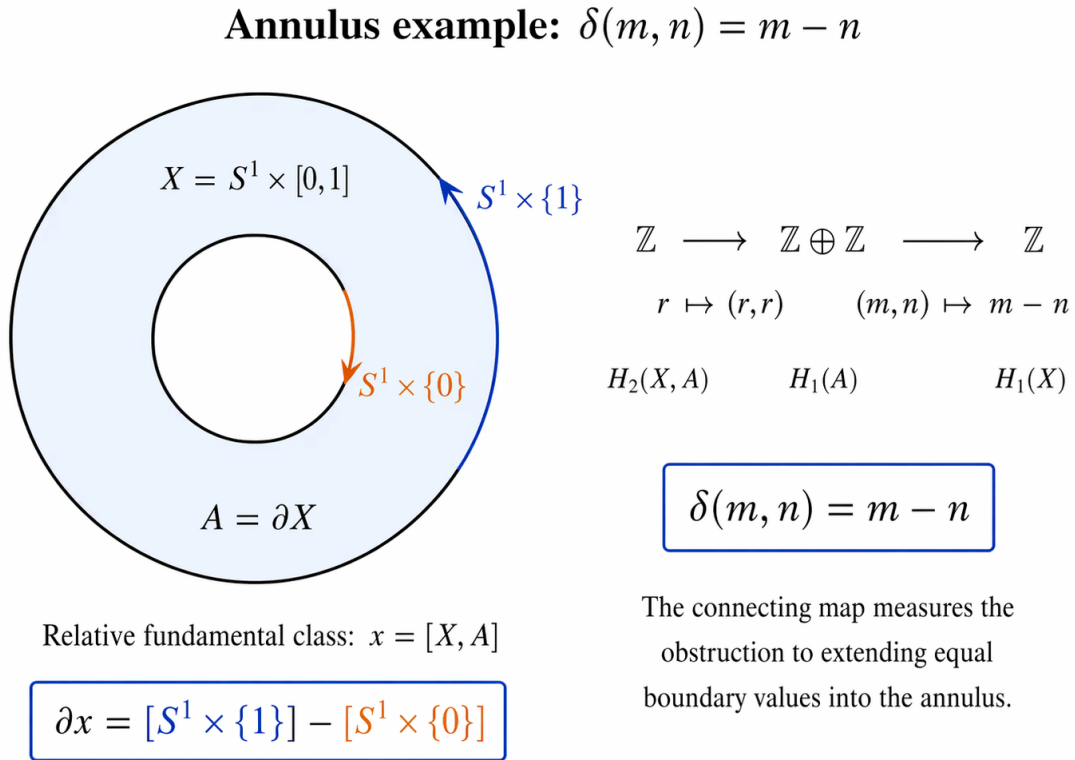


Figure 2: The annulus example. A cohomology class on the boundary has two integer values, one on each boundary component. The connecting map detects their difference: $\delta(m, n) = m - n$.

1.3 The Bockstein homomorphism for $\mathbb{RP}^2$

For the coefficient sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0,$$

the Bockstein homomorphism is

$$\beta : H_q(X; \mathbb{Z}/2) \longrightarrow H_{q-1}(X; \mathbb{Z}).$$

For $X = \mathbb{RP}^2$, the cellular chain complex over $\mathbb{Z}$ contains the boundary map

$$C_2(\mathbb{RP}^2; \mathbb{Z}) \xrightarrow{\times 2} C_1(\mathbb{RP}^2; \mathbb{Z}).$$

Therefore, if $[e^2]$ denotes the mod 2 top cell class, then lifting it to an integral 2-cell gives

$$\partial e^2 = 2e^1.$$

The Bockstein divides this boundary by 2, giving

$$\beta([e^2]) = [e^1].$$

Hence

$$\beta : H_2(\mathbb{RP}^2; \mathbb{Z}/2) \rightarrow H_1(\mathbb{RP}^2; \mathbb{Z})$$

is an isomorphism.

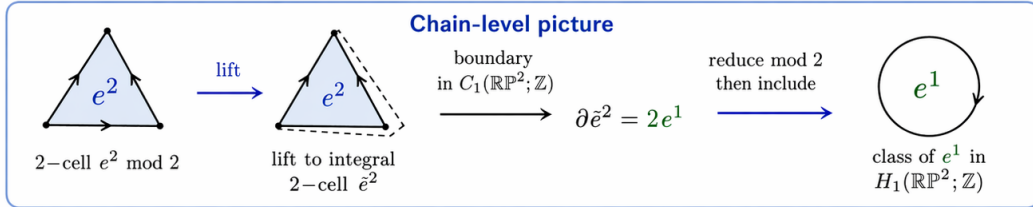
Bockstein for $\mathbb{RP}^2$

**Coefficient
short exact
sequence**

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0$$

**Bockstein
homomorphism**

$$\beta : \underbrace{H_2(\mathbb{RP}^2; \mathbb{Z}/2)}_{\text{generated by } [e^2]} \longrightarrow \underbrace{H_1(\mathbb{RP}^2; \mathbb{Z})}_{\text{generated by } [e^1]}$$



**Integral cellular
chain complex
of $\mathbb{RP}^2$**

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow 0$$

$C_2(\mathbb{RP}^2; \mathbb{Z})$ generated by $\tilde{e}^2$ $C_1(\mathbb{RP}^2; \mathbb{Z})$ generated by e^1

$$H_1(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2$$

generated by $[e^1]$

(since $\ker(\times 2) = \{0\}$, $\text{coker}(\times 2) \cong \mathbb{Z}/2$)

$$\boxed{\beta([e^2]) = [e^1] \quad | \quad \beta \text{ is an isomorphism}}$$

$$H_2(\mathbb{RP}^2; \mathbb{Z}/2) \cong \mathbb{Z}/2 \xrightarrow{\cong} H_1(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2$$

Figure 3: The Bockstein homomorphism for $\mathbb{RP}^2$. The mod 2 top cell detects the integral $\mathbb{Z}/2$ -torsion class one dimension lower.

1.4 Even versus odd real projective spaces

The parity of n matters because

$\mathbb{RP}^n$ is orientable if and only if n is odd.

For even-dimensional projective space,

$$H_{2m}(\mathbb{RP}^{2m}; \mathbb{Z}) = 0,$$

so there is no top-dimensional integral $\mathbb{Z}$ -class. Consequently, the mod 2 action determines the integral action.

For odd-dimensional projective space,

$$H_{2m+1}(\mathbb{RP}^{2m+1}; \mathbb{Z}) \cong \mathbb{Z}.$$

On this top homology group, two maps may differ by multiplication by $+1$ or -1 . But after reducing modulo 2, these two signs become equal:

$$+1 \equiv -1 \pmod{2}.$$

Therefore the converse fails in odd dimensions: the mod 2 homology action does not determine the full integral action.

Even vs odd real projective space

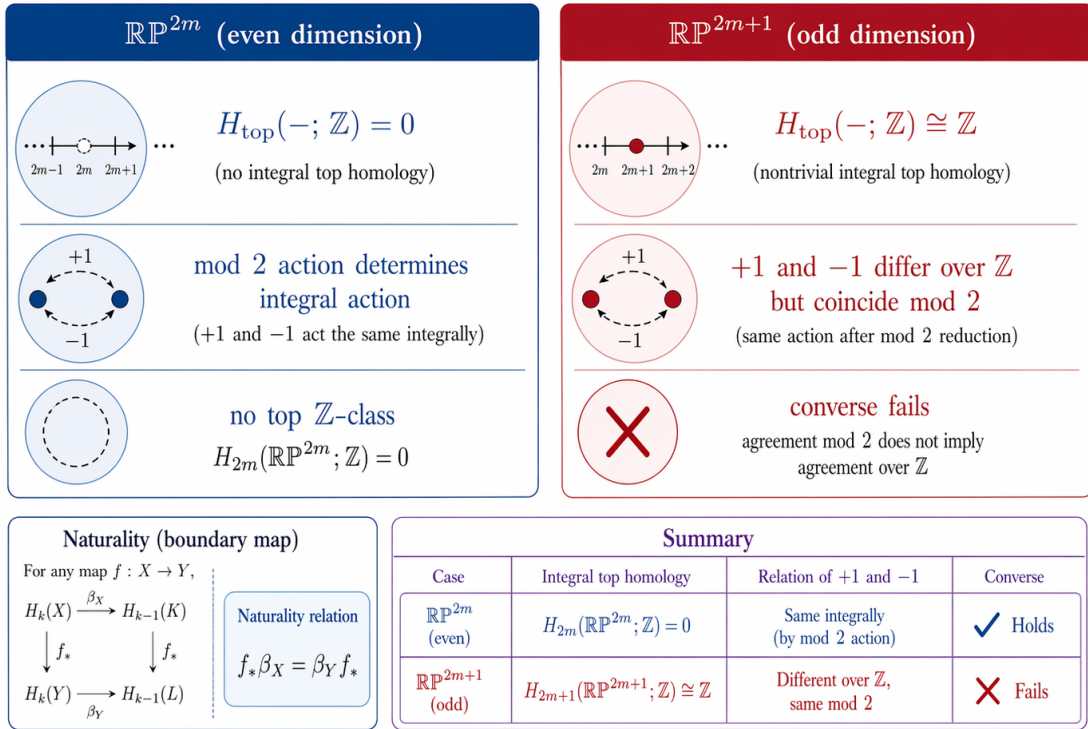


Figure 4: Comparison of even and odd real projective spaces. In even dimensions there is no top integral $\mathbb{Z}$ -class, so the mod 2 action determines the integral action. In odd dimensions, the top integral class is $\mathbb{Z}$, and mod 2 cannot distinguish $+1$ from -1 .

1.5 How to reference the figures in the text

You can refer to the disk–sphere model using Figure 1. The annulus obstruction example is Figure 2. The Bockstein computation for $\mathbb{RP}^2$ is Figure 3. The parity comparison for $\mathbb{RP}^n$ is Figure 4.